

Study Session 7 Checklist

Multicollinearity and VIF (Variance Inflation Factor)

Setup & Encoded Features

- ☐ Open the notebook for Session 7 (Multicollinearity & VIF).
- ☐ Load data/processed/tips_cleaned.csv into df.
- ☐ Create df_encoded = df.copy().
- ☐ Create gender_Male as 1 if gender is male (m/M), else 0.
- ☐ Create is_smoker_Yes as 1 if is_smoker is yes (y/true/1), else 0.
- ☐ Confirm dtypes of gender_Male and is_smoker_Yes are numeric (int).

Select Features for VIF

- ☐ Choose features = [bill_total_usd, party_size, gender_Male, is_smoker_Yes].
- ☐ Check for any missing or non-numeric values in these columns.
- ☐ Optionally drop rows with missing values in the chosen features.

Implement calculate_vif Helper

- ☐ Define calculate_vif(df, features) using statsmodels' variance_inflation_factor.
- ☐ Inside the function: build X = df[features].copy().
- ☐ Add a constant column using statsmodels.tools.add_constant.
- ☐ Compute VIF for each column and return a neat DataFrame.

Compute and Inspect VIF Values

- ☐ Call vif_table = calculate_vif(df_encoded, features).
- ☐ Display the table and confirm VIF values for each feature.
- ☐ Check if any VIF > 5 (moderate) or > 10 (severe).
- ☐ Record your interpretation in Markdown (e.g., no serious issues here).

VIF Visualization

- ☐ Create a bar chart of VIF vs Feature using seaborn or matplotlib.
- ☐ Add horizontal reference lines at VIF=5 and VIF=10 (traffic-light thresholds).
- ☐ Color-code bars (e.g., green <5, yellow 5–10, red >10).
- ☐ Save figure using save_and_log_plot (if available).

Interpretation & Domain Insight

- ☐ Explain why bill_total_usd and party_size have some correlation (but acceptable VIF).
- ☐ Explain why gender_Male and is_smoker_Yes have VIF near 1 (independent).
- ☐ Describe in plain language what multicollinearity does to regression coefficients.
- ☐ Note: High VIF = unstable, noisy, hard-to-interpret coefficients.

Fixing Multicollinearity (Hypothetical Scenarios)

- ☐ Write down at least 3 ways to handle high VIF (e.g., drop, combine, regularize).
- ☐ Give an example of creating a new feature like bill_per_person.
- ☐ Explain when you might use Ridge regression to stabilize coefficients.

Reusable Diagnostics

- ☐ Ensure calculate_vif(df, features) is saved in a helper module or a reusable cell.
- ☐ Add a Markdown note with the VIF interpretation table (1, 5, 10 thresholds).

Study Session 7 Checklist (cont.)

- ☐ Optionally build a small function that prints a VIF warning if any feature > 10 .